

Advanced Statistical Methods II

Practice Problems on Survival Analysis

Fully Worked Solutions with Theory

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This document provides complete, self-explanatory, step-by-step solutions to all twelve practice problems on Survival Analysis. For every problem the full question is quoted first, the exact theory required is recalled, and then each algebraic or conceptual step is justified in detail.

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1 Problem 1: Survival, cumulative hazard, density, and median

Problem Statement

Suppose that the hazard function of a failure time T is

$$h(t) = \begin{cases} 0.05, & 0 < t \leq 6, \\ 0.10, & 6 < t \leq 12, \\ 0.02, & t > 12. \end{cases}$$

1. Find the cumulative hazard $H(t)$.
2. Find the survival function $S(t)$.
3. Find $P(T > 12)$.
4. Find the median survival time.
5. Write the density $f(t)$ in terms of $h(t)$ and $S(t)$.

Required Theory

Recall the fundamental relationships for a continuous non-negative random variable T (event time):

- Hazard function:

$$h(t) = \lim_{\Delta t \downarrow 0} \frac{P(t \leq T < t + \Delta t \mid T \geq t)}{\Delta t} = \frac{f(t)}{S(t)}.$$

- Cumulative hazard:

$$H(t) = \int_0^t h(u) du = -\log S(t) \quad (\text{assuming } S(0) = 1).$$

- Survival function:

$$S(t) = P(T > t) = \exp\{-H(t)\}.$$

- Density:

$$f(t) = h(t)S(t).$$

- Median survival time m is defined by

$$S(m) = \frac{1}{2} \quad (\text{or more precisely } m = \inf\{t : S(t) \leq 1/2\}).$$

These identities are standard; see Kleinbaum & Klein (2012, Ch. 2) or Collett (2015, Ch. 1).

Solution

Step 1 – Cumulative hazard $H(t)$. Because h is piecewise constant we integrate interval by interval.

- For $0 < t \leq 6$:

$$H(t) = \int_0^t 0.05 du = 0.05t.$$

- For $6 < t \leq 12$:

$$H(t) = \int_0^6 0.05 \, du + \int_6^t 0.10 \, du = 0.30 + 0.10(t - 6) = 0.10t - 0.30.$$

- For $t > 12$:

$$H(t) = \int_0^6 0.05 \, du + \int_6^{12} 0.10 \, du + \int_{12}^t 0.02 \, du = 0.30 + 0.60 + 0.02(t - 12) = 0.02t + 0.66.$$

Thus

$$H(t) = \begin{cases} 0.05t, & 0 < t \leq 6, \\ 0.10t - 0.30, & 6 < t \leq 12, \\ 0.02t + 0.66, & t > 12. \end{cases}$$

Step 2 – Survival function $S(t)$. Using $S(t) = \exp\{-H(t)\}$ we obtain

$$S(t) = \begin{cases} \exp(-0.05t), & 0 < t \leq 6, \\ \exp(-(0.10t - 0.30)) = e^{0.30}e^{-0.10t}, & 6 < t \leq 12, \\ \exp(-(0.02t + 0.66)) = e^{-0.66}e^{-0.02t}, & t > 12. \end{cases}$$

Step 3 – $P(T > 12)$.

$$P(T > 12) = S(12) = \exp(-(0.10 \cdot 12 - 0.30)) = \exp(-0.90) = e^{-0.9} \approx 0.4066.$$

Step 4 – Median survival time. We solve $S(m) = 1/2$. First check the interval containing the median:

$$S(6) = e^{-0.30} \approx 0.7408 > 0.5, \quad S(12) = e^{-0.90} \approx 0.4066 < 0.5.$$

Hence $6 < m \leq 12$. In that interval

$$S(m) = e^{0.30}e^{-0.10m} = \frac{1}{2} \Rightarrow e^{-0.10m} = \frac{1}{2}e^{-0.30} \Rightarrow -0.10m = \log \frac{1}{2} - 0.30$$

$$m = \frac{0.30 - \log \frac{1}{2}}{0.10} = \frac{0.30 + 0.693147}{0.10} = 9.93147.$$

So the median is approximately 9.93 months.

Step 5 – Density. By the fundamental identity

$$f(t) = h(t)S(t).$$

Explicitly,

$$f(t) = \begin{cases} 0.05 e^{-0.05t}, & 0 < t \leq 6, \\ 0.10 e^{0.30} e^{-0.10t}, & 6 < t \leq 12, \\ 0.02 e^{-0.66} e^{-0.02t}, & t > 12. \end{cases}$$

Key Takeaway

The piecewise-constant hazard yields a continuous, piecewise-linear cumulative hazard and a continuous, piecewise-exponential survival function. The median is found by solving $S(m) = 1/2$ in the appropriate piece.

2 Problem 2: Risk sets and event times

Problem Statement

The following data contain survival times in months. Here $\Delta = 1$ denotes an observed event and $\Delta = 0$ denotes a right-censored observation.

Subject	1	2	3	4	5	6	7	8	9	10	11	12
Y_i	2	3	4	5	6	7	8	10	11	12	15	16
Δ_i	1	0	1	1	0	1	0	1	0	1	1	0

1. List the distinct event times.
2. For each event time t_j , compute the risk set size n_j and the number of events d_j .
3. Explain why censored observations at earlier times are not included in later risk sets.
4. Compute the Kaplan–Meier estimate $\hat{S}(t)$ at each event time.
5. Estimate the median survival time.

Required Theory

- Distinct ordered event times: $t_{(1)} < t_{(2)} < \cdots < t_{(k)}$ are the unique values of Y_i for which $\Delta_i = 1$.
- Risk set at t_j : $R_j = \{i : Y_i \geq t_j\}$, size $n_j = |R_j|$.
- Number of events at t_j : $d_j = \#\{i : Y_i = t_j, \Delta_i = 1\}$.
- Kaplan–Meier (product-limit) estimator:

$$\hat{S}(t) = \prod_{t_j \leq t} \left(1 - \frac{d_j}{n_j}\right).$$

The curve is a right-continuous step function that jumps only at observed event times.

- Median: $\hat{m} = \inf\{t : \hat{S}(t) \leq 1/2\}$.

(See Kaplan & Meier, 1958; Kleinbaum & Klein, 2012, Ch. 2.)

Solution

Step 1 – Distinct event times. The observed event times ($\Delta_i = 1$) are

$$2, 4, 5, 7, 10, 12, 15.$$

Hence the ordered distinct event times are

$$t_j \in \{2, 4, 5, 7, 10, 12, 15\}.$$

Step 2 – Risk-set sizes and event counts. We list every subject still under observation and event-free just before each t_j .

t_j	Subjects still at risk ($Y_i \geq t_j$)	n_j	d_j
2	all 12 subjects	12	1
4	3–12 (subject 1 failed, subject 2 censored at 3)	10	1
5	4–12	9	1
7	6–12 (subject 5 censored at 6)	7	1
10	8–12 (subject 7 censored at 8)	5	1
12	10–12 (subject 9 censored at 11)	3	1
15	11,12	2	1

Step 3 – Why earlier censored observations leave the risk set. A subject censored at time c contributes the information “ $T > c$ ”. After time c that subject is no longer under observation, therefore cannot experience a later observed event. Consequently the subject is removed from all risk sets R_j with $t_j > c$. This is the definition of the risk set: only individuals still being followed and still event-free are at risk of the next event.

Step 4 – Kaplan–Meier estimates.

$$\begin{aligned}
 \hat{S}(2) &= 1 - \frac{1}{12} = \frac{11}{12} \approx 0.9167, \\
 \hat{S}(4) &= \frac{11}{12} \left(1 - \frac{1}{10}\right) = \frac{11}{12} \cdot \frac{9}{10} = 0.825, \\
 \hat{S}(5) &= 0.825 \cdot \left(1 - \frac{1}{9}\right) = 0.825 \cdot \frac{8}{9} = 0.7333, \\
 \hat{S}(7) &= 0.7333 \cdot \left(1 - \frac{1}{7}\right) = 0.7333 \cdot \frac{6}{7} \approx 0.6286, \\
 \hat{S}(10) &= 0.6286 \cdot \left(1 - \frac{1}{5}\right) = 0.6286 \cdot \frac{4}{5} = 0.5029, \\
 \hat{S}(12) &= 0.5029 \cdot \left(1 - \frac{1}{3}\right) = 0.5029 \cdot \frac{2}{3} \approx 0.3353, \\
 \hat{S}(15) &= 0.3353 \cdot \left(1 - \frac{1}{2}\right) = 0.3353 \cdot \frac{1}{2} \approx 0.1676.
 \end{aligned}$$

Step 5 – Median survival.

$$\hat{S}(10) \approx 0.5029 > 0.5, \quad \hat{S}(12) \approx 0.3353 < 0.5.$$

Hence the estimated median is the first event time at which $\hat{S}$ drops to or below $1/2$, i.e. $\hat{m} = 12$.

Key Takeaway

The KM estimator multiplies successive conditional survival probabilities $1 - d_j/n_j$. Censoring only affects future denominators; it never produces a jump in $\hat{S}$.

3 Problem 3: Likelihood under right censoring

Problem Statement

Let $T_1, \dots, T_n$ be independent continuous event times with density $f(t; \theta)$ and survival function $S(t; \theta)$. We observe (Y_i, Δ_i) , where $Y_i = \min(T_i, C_i)$ and $\Delta_i = I(T_i \leq C_i)$.

1. Derive the likelihood contribution of an individual with an exact observed event time.
2. Derive the likelihood contribution of an individual whose event time is right-censored at y_i .
3. Show that the full likelihood can be written as

$$L(\theta) = \prod_{i=1}^n f(y_i; \theta)^{\Delta_i} S(y_i; \theta)^{1-\Delta_i}.$$

4. Rewrite the likelihood in terms of $h(t; \theta)$ and $H(t; \theta)$.
5. Briefly explain why right censoring is not the same as missing completely at random.

Required Theory

Under independent (non-informative) censoring the contribution of a single observation is

- exact event at y : density $f(y; \theta)$,
- right-censored at y : survival probability $S(y; \theta) = P(T > y)$.

Because $f(t) = h(t)S(t)$ and $S(t) = \exp\{-H(t)\}$, the likelihood admits the equivalent forms

$$L = \prod_i h(y_i)^{\Delta_i} S(y_i) = \prod_i h(y_i)^{\Delta_i} \exp\{-H(y_i)\}.$$

(See Cox & Oakes, 1984; Collett, 2015, Ch. 3.)

Solution

Step 1 – Exact event contribution. If $\Delta_i = 1$ then $Y_i = T_i = y_i$. The probability element of observing an event in $[y_i, y_i + dy]$ is $f(y_i; \theta) dy$. Hence the likelihood contribution is simply

$$L_i(\theta) = f(y_i; \theta).$$

Step 2 – Right-censored contribution. If $\Delta_i = 0$ then $Y_i = C_i = y_i$ and the only information is $T_i > y_i$. Therefore

$$L_i(\theta) = P(T_i > y_i; \theta) = S(y_i; \theta).$$

Step 3 – Full likelihood. Independence of the n subjects yields the product

$$L(\theta) = \prod_{i=1}^n L_i(\theta) = \prod_{i=1}^n f(y_i; \theta)^{\Delta_i} S(y_i; \theta)^{1-\Delta_i}.$$

Step 4 – Hazard form. Substitute $f = hS$:

$$L(\theta) = \prod_{i=1}^n (h(y_i; \theta) S(y_i; \theta))^{\Delta_i} S(y_i; \theta)^{1-\Delta_i} = \prod_{i=1}^n h(y_i; \theta)^{\Delta_i} S(y_i; \theta).$$

Because $S(y) = \exp\{-H(y)\}$ we may also write

$$L(\theta) = \prod_{i=1}^n h(y_i; \theta)^{\Delta_i} \exp\{-H(y_i; \theta)\}.$$

Step 5 – Relation to MCAR. Under MCAR the missingness indicator is independent of both the observed and the unobserved values. Right censoring, however, supplies the inequality $T > Y$. The observed indicator Δ is therefore informative about the magnitude of the unobserved T . Consequently censoring is a structured form of incomplete information, not ordinary MCAR missingness; deleting censored observations would discard genuine partial information and generally bias the estimator of S .

Key Takeaway

The censored likelihood is the product of densities for failures and survival probabilities for censored observations. It is not the same as a complete-data likelihood nor as an MCAR missing-data likelihood.

4 Problem 4: Exponential model with censoring

Problem Statement

Suppose $T \sim \text{Exponential}(\theta)$ with mean parameter $\theta > 0$, so that

$$f(t; \theta) = \theta^{-1} e^{-t/\theta}, \quad S(t; \theta) = e^{-t/\theta}, \quad t > 0.$$

Consider the data

Y_i	4	6	9	13	15	18	20	22
Δ_i	1	0	1	1	0	1	0	1

1. Write the censored likelihood.
2. Derive the MLE of θ .
3. Compute the numerical value of the MLE.
4. Interpret the role of censored observations in the MLE.

Required Theory

For the exponential distribution the hazard is constant: $h(t) = 1/\theta$. The right-censored likelihood reduces to

$$L(\theta) = \theta^{-d} \exp\left(-\frac{1}{\theta} \sum_i Y_i\right),$$

where $d = \sum \Delta_i$ is the number of observed events. The MLE is therefore the total observed time divided by the number of events:

$$\hat{\theta} = \frac{\sum_i Y_i}{d}.$$

(This is the classic result of Epstein & Sobel, 1953; see also Lawless, 2003.)

Solution

Step 1 – Censored likelihood. Number of events $d = \sum \Delta_i = 5$ (subjects with $\Delta = 1$). Total observed time $\sum Y_i = 4 + 6 + 9 + 13 + 15 + 18 + 20 + 22 = 107$. Hence

$$L(\theta) = \theta^{-5} \exp(-107/\theta).$$

Step 2 – MLE. Log-likelihood:

$$\ell(\theta) = -5 \log \theta - 107/\theta.$$

Score equation:

$$\frac{d\ell}{d\theta} = -\frac{5}{\theta} + \frac{107}{\theta^2} = 0 \quad \Rightarrow \quad \hat{\theta} = \frac{107}{5} = 21.4.$$

Second derivative $-5/\theta^2 < 0$ confirms a maximum.

Step 3 – Numerical value.

$$\hat{\theta} = 21.4.$$

Step 4 – Role of censored observations. Every observation, censored or not, contributes its follow-up time Y_i to the denominator of $\hat{\theta}$. A censored observation at y increases the total exposure time without increasing the event count d . Consequently the estimated mean becomes larger (hazard smaller) than if the same subject had failed at y . Ignoring the censored times would therefore under-estimate the mean survival time.

Key Takeaway

Under exponential censoring the MLE is simply total person-time divided by the number of events. Censored subjects lengthen the denominator and correctly pull the mean upward.

5 Problem 5: Weibull model under right censoring

Problem Statement

Let T follow a Weibull distribution with shape $\alpha > 0$ and scale $\lambda > 0$:

$$S(t; \alpha, \lambda) = \exp\left\{-\left(\frac{t}{\lambda}\right)^\alpha\right\}, \quad f(t; \alpha, \lambda) = \frac{\alpha}{\lambda} \left(\frac{t}{\lambda}\right)^{\alpha-1} \exp\left\{-\left(\frac{t}{\lambda}\right)^\alpha\right\}.$$

Use the same data as in Problem 4.

1. Write the censored log-likelihood $\ell(\alpha, \lambda)$.
2. Derive the score equation for λ when α is fixed.
3. Show that the Weibull model reduces to the exponential mean model when $\alpha = 1$ and $\lambda = \theta$.
4. Write R code to maximize the log-likelihood numerically using `optim`.

Required Theory

The Weibull hazard is $h(t) = \frac{\alpha}{\lambda} (t/\lambda)^{\alpha-1}$. The right-censored log-likelihood is

$$\ell(\alpha, \lambda) = \sum_i \left[\Delta_i \log h(Y_i) + \log S(Y_i) \right].$$

When $\alpha = 1$ the Weibull collapses to the exponential distribution with mean λ . (See Collett, 2015, Ch. 6; Kalbfleisch & Prentice, 2002.)

Solution

Step 1 – Censored log-likelihood.

$$\log h(y) = \log \alpha - \log \lambda + (\alpha - 1) \log(y/\lambda), \quad \log S(y) = -(y/\lambda)^\alpha.$$

Hence

$$\begin{aligned} \ell(\alpha, \lambda) &= \sum_i \Delta_i \left[\log \alpha - \log \lambda + (\alpha - 1) \log(Y_i/\lambda) \right] - \sum_i \left(\frac{Y_i}{\lambda} \right)^\alpha \\ &= d \log \alpha - d \log \lambda + (\alpha - 1) \sum_i \Delta_i \log Y_i - (\alpha - 1) d \log \lambda - \lambda^{-\alpha} \sum_i Y_i^\alpha, \end{aligned}$$

where $d = \sum \Delta_i = 5$.

Step 2 – Score for λ (fixed α). Differentiate with respect to λ :

$$\frac{\partial \ell}{\partial \lambda} = -\frac{d}{\lambda} - (\alpha - 1) \frac{d}{\lambda} + \alpha \lambda^{-\alpha-1} \sum_i Y_i^\alpha = \frac{-d\alpha}{\lambda} + \alpha \lambda^{-\alpha-1} \sum_i Y_i^\alpha.$$

Set the derivative to zero:

$$\frac{d\alpha}{\lambda} = \alpha \lambda^{-\alpha-1} \sum_i Y_i^\alpha \quad \Rightarrow \quad \hat{\lambda}^\alpha = \frac{\sum_i Y_i^\alpha}{d}.$$

Thus, for fixed α ,

$$\hat{\lambda}(\alpha) = \left(\frac{\sum_i Y_i^\alpha}{d} \right)^{1/\alpha}.$$

Step 3 – Reduction to exponential. Set $\alpha = 1$. Then

$$S(t) = \exp(-t/\lambda), \quad f(t) = \frac{1}{\lambda} \exp(-t/\lambda),$$

which is exactly the exponential density with mean $\theta = \lambda$. The MLE formula of Step 2 becomes $\hat{\lambda} = \sum Y_i/d$, recovering the result of Problem 4.

Step 4 – R code.

```
# Data from Problem 4
y <- c(4,6,9,13,15,18,20,22)
delta <- c(1,0,1,1,0,1,0,1)

# Negative log-likelihood (to be minimized)
nll <- function(par) {
  alpha <- par[1]; lambda <- par[2]
  if (alpha <= 0 || lambda <= 0) return(1e10)
  d <- sum(delta)
  ell <- d*log(alpha) - d*log(lambda) +
    (alpha-1)*sum(delta*log(y)) -
    (alpha-1)*d*log(lambda) -
    sum((y/lambda)^alpha)
  -ell
}

# Numerical maximization
fit <- optim(c(1.2,20), nll, method="L-BFGS-B",
  lower=c(1e-5,1e-5))
fit$par # approximate MLE (alpha, lambda)
```

Key Takeaway

The Weibull censored likelihood is a straightforward extension of the exponential likelihood; the scale parameter admits a closed-form update for fixed shape, while the shape itself must be obtained numerically.

6 Problem 6: Kaplan–Meier estimator and Greenwood’s formula

Problem Statement

Use the data in Problem 2.

1. Construct the Kaplan–Meier table with columns t_j , n_j , d_j , $\hat{S}(t_j)$.
2. Compute Greenwood’s estimated variance

$$\widehat{\text{Var}}\{\hat{S}(t)\} = \hat{S}(t)^2 \sum_{t_j \leq t} \frac{d_j}{n_j(n_j - d_j)}.$$

3. Give an approximate pointwise 95% confidence interval for $S(10)$ using the plain normal approximation.
4. Explain one limitation of the plain normal interval for survival probabilities.

Required Theory

Greenwood’s formula is the large-sample variance estimator obtained by treating the successive conditional survival probabilities as independent binomial proportions and applying the delta method to the product-limit representation (Greenwood, 1926; Kaplan & Meier, 1958). The plain normal interval $\hat{S} \pm 1.96\sqrt{\widehat{\text{Var}}}$ can produce limits outside $[0, 1]$ and is especially unreliable in the tails when risk sets are small.

Solution

Step 1 – KM table. From the calculations already performed in Problem 2:

t_j	n_j	d_j	$\hat{S}(t_j)$
2	12	1	$11/12 \approx 0.9167$
4	10	1	0.8250
5	9	1	0.7333
7	7	1	0.6286
10	5	1	0.5029
12	3	1	0.3353
15	2	1	0.1676

Step 2 – Greenwood variances. We accumulate the sum $s(t) = \sum_{t_j \leq t} d_j / (n_j(n_j - d_j))$.

$$\begin{aligned}
s(2) &= \frac{1}{12 \cdot 11} = \frac{1}{132} \approx 0.007576, \\
s(4) &= 0.007576 + \frac{1}{10 \cdot 9} = 0.007576 + 0.011111 = 0.018687, \\
s(5) &= 0.018687 + \frac{1}{9 \cdot 8} = 0.018687 + 0.013889 = 0.032576, \\
s(7) &= 0.032576 + \frac{1}{7 \cdot 6} = 0.032576 + 0.023810 = 0.056386, \\
s(10) &= 0.056386 + \frac{1}{5 \cdot 4} = 0.056386 + 0.050000 = 0.106386, \\
s(12) &= 0.106386 + \frac{1}{3 \cdot 2} = 0.106386 + 0.166667 = 0.273053, \\
s(15) &= 0.273053 + \frac{1}{2 \cdot 1} = 0.273053 + 0.500000 = 0.773053.
\end{aligned}$$

Then

$$\begin{aligned}
\widehat{\text{Var}}\{\hat{S}(2)\} &= (0.9167)^2 \cdot 0.007576 \approx 0.00637, \\
\widehat{\text{Var}}\{\hat{S}(10)\} &= (0.5029)^2 \cdot 0.106386 \approx 0.02690, \\
\widehat{\text{Var}}\{\hat{S}(15)\} &= (0.1676)^2 \cdot 0.773053 \approx 0.02172.
\end{aligned}$$

Step 3 – 95% CI for $S(10)$.

$$\sqrt{\widehat{\text{Var}}\{\hat{S}(10)\}} \approx \sqrt{0.02690} \approx 0.1640.$$

$$\hat{S}(10) \pm 1.96 \cdot 0.1640 \approx 0.5029 \pm 0.3214 \quad \Rightarrow \quad [0.1815, 0.8243].$$

Step 4 – Limitation of the plain normal interval. Because $\hat{S}(t)$ is bounded in $[0, 1]$, a symmetric normal interval can easily produce a lower limit below 0 or an upper limit above 1 (especially when $\hat{S}$ is near 0 or 1, or when the variance is large). Moreover the sampling distribution of $\hat{S}(t)$ is typically skewed for small risk sets; the normal approximation therefore under-covers in the tails. Better alternatives are the log-log (complementary log-log) transformation or the arcsine-square-root transformation.

Key Takeaway

Greenwood's formula is easy to compute from the same quantities already needed for the KM estimator, but the resulting normal confidence intervals should be used with caution near the boundaries.

7 Problem 7: Kaplan–Meier as a diagnostic for parametric fit

Problem Statement

For the data in Problem 4, suppose one wants to check whether an exponential model is plausible.

1. Fit the exponential model using the MLE from Problem 4 and write the fitted survival curve.
2. Describe how you would compare the fitted exponential curve with the Kaplan–Meier estimate.
3. Show why, under an exponential model, $-\log S(t)$ should be approximately linear in t through the origin.
4. Show why, under a Weibull model, $\log\{-\log S(t)\}$ should be approximately linear in $\log t$.
5. Write R code to make these two diagnostic plots using Kaplan–Meier estimates.

Required Theory

Graphical goodness-of-fit for parametric survival models rests on the transformations:

- Exponential: $-\log S(t) = \lambda t$ (straight line through origin).
- Weibull: $\log\{-\log S(t)\} = \alpha \log t - \alpha \log \lambda$ (straight line versus $\log t$).

Substituting the nonparametric KM estimate $\hat{S}(t)$ yields diagnostic plots (Collett, 2015, Ch. 4).

Solution

Step 1 – Fitted exponential curve. From Problem 4, $\hat{\theta} = 21.4$, so

$$\hat{S}_{\text{exp}}(t) = \exp(-t/21.4).$$

Step 2 – Comparison with KM. Overlay the continuous exponential curve $\hat{S}_{\text{exp}}(t)$ on the step-function KM estimate computed from the same eight observations. Systematic departure (e.g. the KM curve falling faster or slower than the exponential) indicates lack of fit.

Step 3 – Exponential diagnostic. Under $S(t) = e^{-\lambda t}$ we have

$$-\log S(t) = \lambda t,$$

a straight line through the origin with slope λ . Plotting $-\log \hat{S}_{\text{KM}}(t)$ against t should therefore look roughly linear if the exponential model is adequate.

Step 4 – Weibull diagnostic. Under $S(t) = \exp\{-(t/\lambda)^\alpha\}$ we have

$$-\log S(t) = (t/\lambda)^\alpha, \quad \log\{-\log S(t)\} = \alpha \log t - \alpha \log \lambda.$$

Hence a plot of $\log\{-\log \hat{S}_{\text{KM}}(t)\}$ versus $\log t$ should be approximately linear with slope equal to the shape parameter α .

Step 5 – R code.

```
library(survival)
y <- c(4,6,9,13,15,18,20,22)
delta <- c(1,0,1,1,0,1,0,1)
surv <- Surv(y,delta)
km <- survfit(surv~1)

# Extract KM jumps
t.km <- km$time
S.km <- km$surv

# Exponential diagnostic
plot(t.km, -log(S.km), type="s",
      xlab="t", ylab="-log S(t)",
      main="Exponential diagnostic")
abline(0,1/21.4,col="red") # fitted exponential

# Weibull diagnostic
plot(log(t.km), log(-log(S.km)), type="s",
      xlab="log t", ylab="log(-log S(t))",
      main="Weibull diagnostic")
```

Key Takeaway

The transformations $-\log S$ and $\log(-\log S)$ turn the parametric survival functions into straight lines; the KM estimate supplies a nonparametric benchmark against which linearity (and therefore model adequacy) can be judged visually.

8 Problem 8: Cox partial likelihood by hand

Problem Statement

Consider six subjects with one binary covariate Z .

Subject	A	B	C	D	E	F
Y_i	2	3	5	6	7	8
Δ_i	1	1	0	1	1	0
Z_i	1	0	1	1	0	0

Assume the Cox proportional hazards model

$$h(t | Z) = h_0(t) \exp(\beta Z).$$

There are no tied event times.

1. List the risk set at each event time.
2. Write the Cox partial likelihood $L_p(\beta)$ in this problem.
3. Write the log-partial likelihood $\ell_p(\beta)$.
4. Derive the score equation.
5. Numerically, the solution is approximately $\hat{\beta} = 1.073$. Interpret the hazard ratio.

Required Theory

Under the Cox model the partial likelihood for a single binary covariate (no ties) is

$$L_p(\beta) = \prod_{j: \Delta_j=1} \frac{\exp(\beta Z_j)}{\sum_{i \in R_j} \exp(\beta Z_i)},$$

where R_j is the risk set at the j -th event time (Cox, 1972). The score equation is obtained by differentiating the log-partial likelihood; the MLE $\hat{\beta}$ yields the estimated hazard ratio $e^{\hat{\beta}}$.

Solution

Step 1 – Risk sets at event times. Event times (and corresponding subjects) are 2 (A), 3 (B), 6 (D), 7 (E).

Event time	Subject who fails	Risk set
2	A ($Z = 1$)	{A,B,C,D,E,F}
3	B ($Z = 0$)	{B,C,D,E,F}
6	D ($Z = 1$)	{D,E,F}
7	E ($Z = 0$)	{E,F}

Step 2 – Partial likelihood.

$$L_p(\beta) = \frac{e^\beta}{e^\beta + e^0 + e^\beta + e^\beta + e^0 + e^0} \cdot \frac{e^0}{e^0 + e^\beta + e^\beta + e^0 + e^0} \cdot \frac{e^\beta}{e^\beta + e^0 + e^0} \cdot \frac{e^0}{e^0 + e^0}.$$

Simplifying the denominators:

$$L_p(\beta) = \frac{e^\beta}{3e^\beta + 3} \cdot \frac{1}{2e^\beta + 3} \cdot \frac{e^\beta}{e^\beta + 2} \cdot \frac{1}{2}.$$

Step 3 – Log-partial likelihood.

$$\ell_p(\beta) = \beta - \log(3e^\beta + 3) - \log(2e^\beta + 3) + \beta - \log(e^\beta + 2) - \log 2.$$

Step 4 – Score equation. Differentiate:

$$\begin{aligned} \frac{d\ell_p}{d\beta} &= 1 - \frac{3e^\beta}{3e^\beta + 3} - \frac{2e^\beta}{2e^\beta + 3} + 1 - \frac{e^\beta}{e^\beta + 2} \\ &= 2 - \frac{e^\beta}{e^\beta + 1} - \frac{2e^\beta}{2e^\beta + 3} - \frac{e^\beta}{e^\beta + 2}. \end{aligned}$$

Set the derivative equal to zero and solve numerically; the unique root is $\hat{\beta} \approx 1.073$.

Step 5 – Hazard-ratio interpretation.

$$\widehat{\text{HR}} = e^{\hat{\beta}} \approx e^{1.073} \approx 2.92.$$

Subjects with $Z = 1$ have an estimated instantaneous hazard approximately 2.92 times that of subjects with $Z = 0$, at any fixed time, under the proportional-hazards assumption.

Key Takeaway

The Cox partial likelihood is formed solely from the relative risks of the subjects who fail versus those still at risk; the baseline hazard cancels and never needs to be estimated.

9 Problem 9: Hazard ratios and survival curves under proportional hazards

Problem Statement

Suppose the Cox model with a single continuous covariate X is

$$h(t | X) = h_0(t) \exp(0.35X).$$

1. Interpret the hazard ratio for a one-unit increase in X .
2. Interpret the hazard ratio for a five-unit increase in X .
3. If $S_0(t) = \exp\{-H_0(t)\}$ denotes the baseline survival curve, show that

$$S(t | X) = S_0(t)^{\exp(0.35X)}.$$

4. If $S_0(24) = 0.70$, compute $S(24 | X = 2)$.
5. Explain why the proportional hazards assumption is about hazards, not survival probabilities.

Required Theory

Under the proportional-hazards model

$$h(t | X) = h_0(t)e^{\beta X} \Rightarrow H(t | X) = H_0(t)e^{\beta X} \Rightarrow S(t | X) = \exp\{-H_0(t)e^{\beta X}\} = S_0(t)^{e^{\beta X}}.$$

The hazard ratio $e^{\beta \Delta x}$ is constant over time; the survival curves themselves are not parallel on the original scale.

Solution

Step 1 – One-unit increase.

$$\frac{h(t | X + 1)}{h(t | X)} = e^{0.35} \approx 1.419.$$

A one-unit increase in X multiplies the instantaneous hazard by approximately 1.42 at every t .

Step 2 – Five-unit increase.

$$\frac{h(t | X + 5)}{h(t | X)} = e^{0.35 \cdot 5} = e^{1.75} \approx 5.755.$$

A five-unit increase multiplies the hazard by roughly 5.76.

Step 3 – Survival formula.

$$H(t | X) = \int_0^t h(u | X) du = H_0(t)e^{0.35X}.$$

$$S(t | X) = \exp\{-H(t | X)\} = \exp\{-H_0(t)e^{0.35X}\} = [\exp\{-H_0(t)\}]^{e^{0.35X}} = S_0(t)^{e^{0.35X}}.$$

Step 4 – Numerical evaluation.

$$S(24 | X = 2) = 0.70^{e^{0.70}} = 0.70^{e^{0.7}} \approx 0.70^{2.0138} \approx 0.489.$$

Step 5 – PH is about hazards. The assumption asserts that the ratio of hazards is constant over time. The survival curves $S(t | X)$ are powers of the baseline survival; they are therefore not parallel on the probability scale, nor do they maintain a constant vertical distance. Only after a complementary-log-log transformation do the curves become parallel (the vertical distance being exactly βX).

Key Takeaway

Hazard ratios are multiplicative and constant; survival curves are related by a power transformation and therefore change shape with the covariate.

10 Problem 10: R exercise – Kaplan–Meier, log-rank, Cox

Problem Statement

This exercise uses the `lung` dataset from the R package `survival`.

1. Load the data and recode the status variable so that 1 denotes death and 0 denotes censoring.
2. Fit a Kaplan–Meier curve stratified by sex.
3. Estimate the median survival time in each group.
4. Perform a log-rank test comparing the two groups.
5. Fit a Cox model using sex, age, and ECOG performance score as covariates.
6. Interpret the coefficient of sex and the coefficient of age.

Required Theory

The `survival` package implements the product-limit estimator (`survfit`), the log-rank test (`survdif`), and the Cox partial-likelihood maximizer (`coxph`). Status coding must be 0 = censored, 1 = event for the majority of functions.

Solution (complete R script with commentary)

```
library(survival)
library(survminer)  # optional, for nicer plots

# 1. Load and recode
data(lung)
# original status: 1=censored, 2=dead
lung$status01 <- as.integer(lung$status==2)  # 1=death, 0=censor

# 2. KM stratified by sex
km.sex <- survfit(Surv(time,status01)~sex, data=lung)
print(km.sex)
# plot(km.sex, col=c(1,2), xlab="Days", ylab="Survival")

# 3. Median survival
# the print method already shows the medians;
# explicit extraction:
quantile(km.sex, probs=0.5)

# 4. Log-rank test
survdif(Surv(time,status01)~sex, data=lung)

# 5. Cox model
cox.fit <- coxph(Surv(time,status01)~sex+age+ph.ecog, data=lung)
summary(cox.fit)

# 6. Interpretation
# sex (female=2 vs male=1): exp(coef) is the hazard ratio
# HR < 1 means lower hazard (better survival) for females
```

```
# age: exp(coef) is the multiplicative change in hazard  
#   for a one-year increase in age, holding other covariates fixed
```

Typical numerical results (exact numbers depend on the version of the package):

- Median survival is longer for females than for males.
- The log-rank test yields a highly significant p -value, confirming a difference between the two survival curves.
- In the Cox model the coefficient of sex is negative ($HR < 1$), indicating a protective effect of being female; the coefficient of age is positive ($HR > 1$), indicating that older patients have higher hazard.

Key Takeaway

A complete applied survival analysis in R consists of (i) correct status coding, (ii) nonparametric description (KM + log-rank), and (iii) a semi-parametric regression model (Cox) with clear hazard-ratio interpretations.

11 Problem 11: R exercise – simulation under a Cox–Weibull model

Problem Statement

Generate data from a proportional hazards model with a Weibull baseline and one binary covariate.

1. Use the supplied code skeleton to simulate the data.
2. Fit a Kaplan–Meier curve by z .
3. Fit a Cox model with z as the covariate.
4. Compare the estimated hazard ratio with the true hazard ratio.
5. Create a plot of $\log\{-\log \hat{S}(t)\}$ versus $\log t$ for the two groups and explain what approximately parallel curves would suggest.

Required Theory

The inverse-transform method for a Weibull PH model with shape α and scale λ yields

$$T = \lambda(-\log U/e^{\beta z})^{1/\alpha}, \quad U \sim \text{Unif}(0, 1).$$

Under the true PH model the estimated Cox hazard ratio should recover e^β , and the complementary-log-log survival curves should be parallel with vertical separation β .

Solution

Step 1 – Simulation (given skeleton).

```
set.seed(202627)
n <- 250
z <- rbinom(n, size=1, prob=0.5)
beta <- log(1.8)
alpha <- 1.4
lambda <- 12
u <- runif(n)
Ttrue <- lambda * (-log(u)/exp(beta*z))^(1/alpha)
C <- rexp(n, rate=0.035)
time <- pmin(Ttrue,C)
status <- as.integer(Ttrue<=C)
dat <- data.frame(time=time, status=status, z=z)
```

Steps 2–4 – KM, Cox, comparison.

```
library(survival)
km.z <- survfit(Surv(time,status)~z, data=dat)
print(km.z)

cox.z <- coxph(Surv(time,status)~z, data=dat)
summary(cox.z)
# estimated HR should be close to the true value 1.8
```

Step 5 – Complementary-log-log plot.

Extract KM for each group

```
plot(km.z, fun="cloglog",  
      xlab="log t", ylab="log(-log S(t))",  
      col=c(1,2), main="cloglog diagnostic")
```

Approximately parallel curves indicate that the proportional-hazards assumption holds (the vertical distance estimates the log-hazard ratio β). Any systematic change in slope between the two groups would signal a violation of PH.

Key Takeaway

A correctly specified Cox–Weibull simulation recovers both the parametric hazard ratio and the nonparametric parallelism of the cloglog plots, confirming that the data-generating mechanism is consistent with the modelling assumptions.

12 Problem 12: Conceptual question on informative censoring

Problem Statement

In a clinical study, patients are followed until relapse. Some patients are censored because the study ends. Others are censored because their health deteriorates and they stop attending follow-up visits.

1. Which censoring mechanism is more likely to be independent/non-informative?
2. Why can informative censoring bias the Kaplan–Meier estimator?
3. Give one practical strategy for detecting or reducing the impact of informative censoring.
4. Explain the connection with missing-data mechanisms such as MCAR, MAR, and MNAR.

Required Theory

Independent (non-informative) censoring means that, conditional on the observed covariates (if any), the censoring time C is independent of the event time T . When this fails, the risk sets become selectively depleted of high-risk (or low-risk) subjects and the product-limit estimator is biased. The analogy with missing-data mechanisms is exact: independent censoring corresponds to MAR (or MCAR), while informative censoring corresponds to MNAR.

Solution

Step 1 – Which mechanism is independent? Administrative end-of-study censoring is determined by the calendar date of analysis and is therefore independent of each patient's latent relapse time (given any measured covariates). Censoring caused by deteriorating health is typically dependent on the same latent process that drives relapse; it is therefore informative.

Step 2 – Why bias arises. Under informative censoring the subjects who remain under observation are no longer a random sample of those who would still be event-free. If sicker patients drop out, the risk sets become depleted of high-risk individuals; the KM estimator then over-estimates survival. Conversely, if healthier patients are lost, survival is under-estimated. Because the KM formula uses only the observed risk-set sizes, it cannot correct for this selection.

Step 3 – Practical strategies.

- Collect and model the reasons for dropout (e.g., a competing-risk or joint longitudinal-survival model).
- Perform sensitivity analyses that impute a range of plausible residual lifetimes for the informatively censored subjects.
- Use inverse-probability-of-censoring weights (IPCW) when a model for the censoring hazard can be built from observed covariates.

Step 4 – Link to missing-data language.

Censoring language	Missing-data language
Independent / non-informative	MAR (or MCAR if no covariates)
Informative	MNAR

In both frameworks the key question is whether the mechanism that produces the incomplete observation depends on the unobserved value itself. When it does, the ordinary observed-data likelihood (or the ordinary KM product) is no longer valid without additional modelling of that mechanism.

Key Takeaway

Administrative censoring is usually safe; dropout related to the patient's evolving health is not. The same conceptual distinction that appears in missing-data theory (MAR versus MNAR) reappears in survival analysis as independent versus informative censoring.

A Notation Quick Reference

Symbol	Meaning
T, C, Y, Δ	event time, censoring time, $\min(T, C)$, event indicator
$S(t), f(t), h(t), H(t)$	survival, density, hazard, cumulative hazard
n_j, d_j	risk-set size and event count at t_j
$\hat{S}(t)$	Kaplan–Meier estimator
$\ell(\theta), L(\theta)$	log-likelihood / likelihood
$L_p(\beta)$	Cox partial likelihood

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